

CUSTOMER CHURN PREDICTION PIPELINE

DEPI Graduation Project Documentation

Full Technical Documentation — All Roles & Deliverables

Team: Saliha 3la Allah

Python 3.14.1 | LightGBM | Microsoft SQL Server | Streamlit | Power BI

End-to-End Machine Learning Pipeline for Telecom Customer Retention

1. Project Overview

This project builds a complete end-to-end Machine Learning pipeline to predict customer churn for a telecom company. The goal is to identify customers likely to leave before they do, enabling proactive retention strategies. The final deliverable includes a Streamlit prediction & retention application and a Power BI business intelligence dashboard covering revenue impact, customer segmentation, and behavioral risk signals.

Property	Value
Dataset	customer_churn.csv
Total Customers	999,999
Features	32 columns
Target Variable	churn (0 = Stayed, 1 = Left)
Churn Rate	9.92%
Retention Rate	90.08%
Best Model	LightGBM
F1-Score (Churn)	0.257
Language	Python 3.14.1
Database	Microsoft SQL Server 16.0
Dashboard (App)	Streamlit
Dashboard (BI)	Power BI
Team Name	Saliha 3la Allah

2. Project Structure

Path	Description
data/raw/customer_churn.csv	Original raw dataset — never modified
data/processed/clean_data.csv	Cleaned, encoded, scaled dataset
notebooks/01_data_preprocessing.ipynb	Initial data exploration and full preprocessing pipeline
notebooks/02_model_training.ipynb	Model training & comparison
notebooks/03_model_evaluation.ipynb	Metrics & confusion matrix
notebooks/04_sql_analysis.ipynb	SQL queries & integration
database/schema.sql	SQL Server table schema
database/load_data.py	Load CSV to SQL Server
database/queries/	5 analytical SQL queries
models/lgbm_model.pkl	Trained LightGBM model
models/scaler/scaler.pkl	Fitted StandardScaler
app/streamlit_app.py	Interactive dashboard (Overview, EDA, Prediction, Retention Plan)

Path	Description
dashboard/churn_dashboard.pbix	Power BI business intelligence dashboard
reports/	5 PDF reports
requirements.txt	Python dependencies
README.md	Project overview

3. Team Roles & Deliverables

Team Name: Saliha 3la Allah

Role	Responsibilities	Deliverables
Data Engineer	Cleaning, encoding, scaling, pipeline	clean_data.csv, preprocessing notebook
Data Scientist	Model training, evaluation, selection	lgbm_model.pkl, training notebooks
Data Analyst	EDA, visualizations, insights	EDA notebook, eda_report.pdf
SQL Engineer	Database design, queries, analysis	schema.sql, 5 queries, sql_report.pdf
BI Developer	Power BI dashboard (Executive Summary, Revenue Impact, Segmentation)	churn_dashboard.pbix
App Developer	Streamlit dashboard, prediction & retention interface	streamlit_app.py

4. Data Engineering Pipeline

The preprocessing pipeline transforms the raw 32-column dataset into a clean, model-ready format through 7 sequential steps.

Step	Action	Result
1	Load raw data	999,999 rows × 32 columns
2	Drop customer_id, signup_date	30 columns remaining
3	Fill missing values with Median	5 columns fixed — 0 nulls
4	Ordinal Encoding	contract, education, paperless_billing
5	One-Hot Encoding	gender, marital_status, payment_method
6	StandardScaler	13 numerical columns scaled
7	Train/Test Split (80/20)	799,999 train 200,000 test

5. Machine Learning Models

Four classification models were trained and evaluated. The primary challenge was class imbalance (9:1 ratio), addressed using class_weight='balanced'.

Model	F1-Score	Recall	Precision	ROC-AUC	Status
LightGBM	0.257	0.636	0.161	0.635	Selected
Logistic Regression	0.256	0.642	0.160	0.635	Runner-up
XGBoost	0.253	0.603	0.160	0.627	Tested
Random Forest	0.001	0.000	0.545	0.500	Needs tuning

Why LightGBM was selected

Best overall F1-Score and ROC-AUC. Significantly faster on 1M+ rows compared to other tree-based models. Production-ready with strong support for class imbalance, and forms the basis for both the real-time Prediction page and the Retention Plan risk-scoring engine (Model Identified recall of 63.6%) in the Streamlit app.

6. SQL Database & Analysis

The dataset was loaded into Microsoft SQL Server for structured querying and business intelligence analysis.

Property	Value
Server	OMAR-ZIZO (SQL Server 16.0.1180)
Database	customer_churn
Table	dbo.customers
Row Count	999,999
Column Count	38 (after encoding)
Queries Executed	5 analytical queries

Key SQL Findings

Query	Finding
Overall Churn Rate	9.92% — Class imbalance of 9:1
Churn by Contract	Month-to-Month: 26.51% One Year: 12.65% Two Year: 5.65%
High-Risk Customers	100 retained customers with high complaints & low satisfaction
Churn by Charges	Minimal variation — pricing not primary driver
Churn by Tenure	New customers (<1yr): 10.26% Loyal (3yr+): 9.33%

7. Key EDA Insights

#	Insight	Business Action
1	New customers (0-10 months) churn most and are the highest-risk window	Launch a 90-day onboarding program with weekly check-ins

#	Insight	Business Action
2	Month-to-Month contracts = 26.5% churn vs 5.6% for Two-Year	Offer discounts to switch to annual/two-year plans
3	High complaints = strong churn signal	Resolve complaints within 24 hours
4	Low satisfaction predicts churn (17% churn at satisfaction 1 vs ~7% at satisfaction 9)	Regular satisfaction surveys + follow-up
5	Very High charges = 12% churn rate vs 8% for Medium charges	Review pricing for high-charge segments
6	Customers with 4+ late payments churn at ~20%, the highest of any group	Proactive billing support / flexible payment plans
7	Customers without tech support churn more (11.28%) than those with it (8.56%)	Bundle free tech support trials for at-risk customers
8	Fewer active services correlates with higher churn	Offer 3+ service bundles at a discount

8. Streamlit Dashboard — "Churn Intelligence"

An interactive web dashboard called "Churn Intelligence" (team: Saliha 3la Allah) was built using Streamlit to visualize insights, run real-time churn predictions, and generate tailored retention plans. The dashboard now spans 4 pages and shows a live "Model Online" status indicator.

Page	Description
Overview	4 metric cards (Total Customers, Churned, Retained, Month-to-Month Churn) + Key Insights summary (Contract Type, New Customers, Complaints, High Monthly Charges)
EDA	Interactive charts — Churn Distribution and Churn Rate by Contract Type (26.3% / 12.4% / 5.6%)
Prediction	Input sliders (Age, Complaints, Tenure, Satisfaction, Monthly Charges, Credit Score, Contract Type, Late Payments) + LightGBM churn probability gauge + risk classification
Retention Plan	NEW — Customer profile inputs (Age, Complaints, Tenure, Satisfaction, Contract Type, Number of Services, Monthly Charges, Late Payments, Credit Score) → "Generate Retention Plan" → churn probability, risk factor scores, and prioritized recommended actions (Urgent / Soon / Normal)

8.1 Overview Page

Key metrics displayed:

- Total Customers: 999,999
- Churned: 99,227 (9.92% churn rate)
- Retained: 900,772 (90.08% retention)
- Month-to-Month Churn: 26.5% — flagged as the highest-risk segment

Key Insights panel: Contract Type is the strongest churn predictor (Month-to-Month churn at 26.5% vs 5.6% for Two-Year); new customers (0–10 months) are at highest risk and churn drops significantly after the first year; complaints are

a strong early warning signal; high monthly charges correlate with higher churn (Very High segment shows 12% vs 8% for Medium).

8.2 EDA Page

Visual breakdown of churn patterns across key features, including a Churn Distribution bar chart (No Churn vs Churn) and a Churn Rate by Contract Type chart showing Month-to-Month at 26.3%, One Year at 12.4%, and Two Year at 5.6%.

8.3 Prediction Page

Users enter a customer profile through interactive sliders and dropdowns, grouped into two sections:

- Customer Profile — Age (18–90), Complaints (0–7), Tenure in Months (0–72), Satisfaction 1–9, Monthly Charges (\$20–120), Credit Score (300–850)
- Account Details — Contract Type (Month-to-Month / One Year / Two Year), Late Payments (0–10)

The model returns a Churn Probability gauge (e.g. 36.7%) with a Low / Medium / High Risk label and a plain-language summary such as "Low Risk — Customer is stable."

8.4 Retention Plan Page (New Feature)

This new page analyzes customer risk and produces a tailored retention strategy using the same LightGBM-based scoring engine.

- Customer Profile inputs: Age, Complaints, Contract Type, Tenure (Months), Satisfaction (1–9), Number of Services (0–8), Monthly Charges (\$20–120), Late Payments (0–10), Credit Score (300–850)
- Churn Probability gauge with a risk band (e.g. 34.6% — Medium Risk), noted as "Based on LightGBM model"
- Risk Factor Scores — individual bars for Complaints, Satisfaction, Tenure Risk, Late Payments, and Contract Risk (e.g. Tenure Risk 91%, Satisfaction 22%, Complaints 14%)
- Recommended Retention Actions with priority tags:
 - New Customer Onboarding (URGENT) — enroll customers within their first 6 months into a 90-day onboarding program with weekly check-ins and a welcome bonus
 - Service Bundle Upsell (NORMAL) — offer a bundle deal with 3+ services at a 15% discount for customers with fewer than 3 active services

A summary strip counts actions by priority (e.g. 1 Urgent · 0 Soon · 1 Normal) with guidance to start with urgent items first.

How to Run

```
cd D:\project cd D:\project streamlit run app/streamlit_app.py
```

9. Power BI Business Intelligence Dashboard

In addition to the Streamlit app, a full Power BI dashboard was built for executive-level and business-stakeholder reporting. It contains six report pages, navigated via a left-hand icon rail: Executive Summary, Revenue & Financial Impact, Contract & Tenure Analysis, Satisfaction & Complaints Deep Dive, Customer Behavior & Risk Signals, and Customer Segmentation.

9.1 Executive Summary

Top-line KPI cards: Total Customers 1M, Churned Customers 99K, Churn Rate 9.92%, Lost Monthly Revenue \$8.46M, Lifetime Revenue Lost \$173.41M, and Average Satisfaction 6.16. Includes a churn-rate gauge (9.92% against a 25% warning threshold), a donut chart of Retained vs Churned customers (90.08% / 9.92%), and a decomposition tree breaking down customer count by contract-related dimensions.

9.2 Revenue & Financial Impact

Metric	Value
Lifetime Revenue Lost	\$173.41M
Lost Monthly Revenue	\$8.46M
Sum of Cumulative Revenue Lost	173.41M
Lost Monthly Revenue — highest by contract	One-Year contract customers

Additional visuals: Sum of Monthly Charges by Customer Status (Retained vs Lost, with an increase/decrease waterfall), a scatter of monthly charges vs total charges by customer status, a top-customers table sorted by total charges lost, and a rising trendline of average cumulative revenue lost by tenure — revenue lost climbs steadily the longer a lost customer had been on the books.

9.3 Contract & Tenure Analysis

Confirms the SQL findings at scale: Churn Rate by Contract is 26.51% (Month-to-Month), 12.65% (One Year), and 5.65% (Two Year). A stacked bar of Total Customers by contract and status shows Month-to-Month has the smallest customer base but the highest loss proportion. Churn Rate by Tenure shows an overall declining trend as tenure increases, with early-tenure months (1–6) consistently above the 9.92% average and long-tenured customers trending below it.

9.4 Satisfaction & Complaints Deep Dive

A cross-tab of churn rate by contract type and number of complaints shows churn compounding sharply for Month-to-Month customers as complaints rise (up to ~100%+ of that segment at the highest complaint counts, vs. much flatter curves for One-Year and Two-Year). Churn Rate by Satisfaction Score declines steadily from roughly 17% at the lowest satisfaction rating to about 7% at the highest, confirming satisfaction as a leading indicator alongside contract type.

9.5 Customer Behavior & Risk Signals

Signal	Finding
High-Risk Retained Customers	380K — retained customers who still show high-risk behavioral signals

Signal	Finding
Complaints by Status	Lost customers account for 12.78% of total complaints vs 87.22% for Retained
Churn by Late Payments	Churn rate rises with late-payment count, peaking at ~20%+ for customers with 4 or more late payments
Churn by Tech Support	11.28% churn without tech support vs 8.56% with tech support
Churn by Number of Services	Churn rate is highest for customers with the fewest active services and decreases as service count increases
Engagement Decay	Monthly charges from retained customers taper off sharply the longer since their last interaction, more so than for lost customers

9.6 Customer Segmentation

Customers are grouped into Low Risk, Medium Risk, and High Risk segments:

- Low Risk: 905K customers (90.46%) — lowest churn rate of the three segments
- High Risk: elevated churn rate, second-largest revenue exposure
- Medium Risk: 93K customers (9.32%) — highest churn rate of all segments

A funnel view quantifies the retention opportunity: Total Risk Revenue of \$6.84M, of which the model identifies \$4.35M (63.6% recall), with an Expected Saved figure of \$2.18M assuming a 50% intervention success rate (31.8% of total risk revenue).

10. Requirements & Installation

Python Version

Python 3.14.1

Install All Dependencies

```
pip install -r requirements.txt
```

Package	Version	Purpose
pandas	2.2.3	Data manipulation
numpy	2.1.3	Numerical computing
scikit-learn	1.5.2	ML pipeline & metrics
lightgbm	4.5.0	Primary ML model
xgboost	2.1.3	Alternative ML model
imbalanced-learn	0.12.4	SMOTE for class imbalance
matplotlib	3.9.3	Data visualization
seaborn	0.13.2	Statistical visualization
pyodbc	5.2.0	SQL Server connection
streamlit	1.40.2	Web dashboard (Overview / EDA / Prediction / Retention Plan)
joblib	1.4.2	Model serialization

Note: The Power BI dashboard (churn_dashboard.pbix) requires Power BI Desktop to open and edit; no additional Python packages are needed to view it, only a connection to database/schema.sql or the exported clean_data.csv.

End of Project Documentation — Team Saliha 3la Allah